
Deep Learning for High-Frequency Bitcoin Limit Order Book Forecasting

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Abstract

We investigate short-horizon mid-price movement prediction on high-frequency Bitcoin perpetual futures limit order book (LOB) data, using a contiguous five-day sample of approximately 1.7 million snapshots. We compare five approaches across a methodological spectrum: an Order Flow Imbalance (OFI) baseline, a plain LSTM, DeepLOB, LiT, and our proposed Attentive DeepLOB¹ extension which augments DeepLOB with two axial self-attention blocks. Performance is evaluated across 1 s, 2 s, and 5 s horizons using accuracy, balanced accuracy, and macro and weighted F1-score under a shared preprocessing, labelling, and day-based chronological splitting pipeline. Our results show that greater architectural complexity does not automatically imply stronger generalisation. The OFI baseline remains a non-trivial benchmark against a 33% chance level. Deeper sequence models substantially outperform OFI, and Attentive DeepLOB achieves the best accuracy at the 1 s horizon (68.5%). The original DeepLOB exhibits fast early overshoot, suggesting its convolutional inductive bias fits the training distribution rapidly but does not generalise as stably as the attention-augmented alternatives. These findings frame deep LOB forecasting as a robustness problem and highlight the trade-off between specialised spatial inductive bias and generalisation on noisy real Bitcoin order book data.

1. Introduction

A limit order book (LOB) is a list of all outstanding buy and sell orders for an asset, organised by price. Buy orders, called bids, sit at prices below the current trading price, and sell orders, called asks, sit above. The best bid is the highest current buy price and the best ask the lowest current sell price. The midpoint between the two is the mid-price, and the difference is the spread, which measures the cost of immediately crossing the book to take liquidity. Forecasting whether the mid-price will move up, move down, or remain

approximately unchanged over the next short horizon is directly relevant to market makers, algorithmic traders, and risk managers.

Cryptocurrency markets, and in particular Bitcoin perpetual futures, provide an especially demanding setting for this forecasting problem. Perpetual futures track the underlying asset price but, unlike standard futures, do not expire. Centralised crypto platforms operate continuously, attract globally distributed order flow, and exhibit high trading activity. At sub-second-to-second timescales, price movements are driven not by macroeconomic news but by microstructure dynamics including the immediate interaction of orders, cancellations, liquidity, and queue imbalances within the order book (Cont et al., 2014). This makes the raw LOB snapshot a natural input for short-horizon forecasting models. The task is conceptually challenging because labels are noisy, classes are heavily imbalanced at sub-second horizons, and naive baselines can already achieve high raw accuracy by predicting the majority class, so meaningful evaluation requires balanced metrics and theoretically grounded baselines.

Recent advances in deep learning have produced architectures specifically designed for LOB data. Zhang et al. (2019) introduced DeepLOB, which exploits the spatial structure of the LOB snapshot via convolutional layers and an Inception module before capturing temporal dynamics with an LSTM, an architecture we describe in detail in Section 4.3. Transformer-based alternatives have followed, replacing locality-biased CNN feature extraction with global self-attention (Berti & Kasneci, 2025; Xiao et al., 2025). A practical concern with the dominant LOB benchmarks is that they are not always representative of live trading conditions.

The widely-used FI-2010 dataset of Ntakaris et al. (2018) on which DeepLOB was originally evaluated provides only ten days of data from five Finnish stocks, is event-sampled at intervals of ten LOB updates rather than uniformly in time, and ships pre-normalised so that the original prices and volumes cannot be recovered. Zhang et al. (2019) themselves describe FI-2010 as “far too short, downsampled and taken from a less liquid market,” and Briola et al. (2025) document that several state-of-the-art models overfit FI-2010 and lose substantial performance on more realistic LOBSTER data.

¹Referred to as DeepLOB v2 in the code and plots.

LiT was evaluated on Binance Bitcoin perpetual futures reconstructed at the millisecond level but then sampled following the same event-based scheme as [Ntakaris et al. \(2018\)](#), recording a new snapshot only when an order book event occurs. Both LOB architectures have therefore been benchmarked on event-sampled data and never directly compared on a single shared dataset. Event-based sampling is convenient because it concentrates the data on moments where something is happening, but this also makes the prediction task easier in a way that does not transfer to live trading: the model never has to handle quiet stretches where the book is stable, the time interval between consecutive samples becomes a function of market activity that the model can implicitly exploit, and prediction horizons defined in milliseconds (such as the 300–500 ms windows used by [Xiao et al. \(2025\)](#)) actually correspond to “the next event after this many milliseconds have elapsed,” which is not the same target as predicting the price one second from now in clock time.

We therefore test DeepLOB and LiT on the same uniform 250 ms-sampled BTCUSDT.P stream without any benchmark-specific filtering, normalisation, or event-thinning. This is a more robust evaluation setting because every quiet stretch is preserved, the time interval between consecutive snapshots is fixed and independent of activity, and prediction horizons in seconds correspond to clock time as a trader would experience it. Both DeepLOB and LiT face the same task that practitioners actually face. We then propose our own architecture in this same setting.

We conduct a controlled comparative study of five models. The OFI baseline of [Cont et al. \(2014\)](#) provides a theory-grounded non-machine-learning benchmark. A plain LSTM acts as an ablation of the CNN front-end. DeepLOB ([Zhang et al., 2019](#)) is the purpose-built CNN-Inception-LSTM architecture for LOB data. LiT ([Xiao et al., 2025](#)) is a Vision-Transformer-inspired LOB architecture with a structured patch embedding and an LSTM head. Our proposed Attentive DeepLOB inserts two axial self-attention blocks between DeepLOB’s Inception module and LSTM, motivated by the hypothesis that DeepLOB’s strong but local convolutional bias may be too rigid for noisy crypto data, and that adding global attention without abandoning the CNN front-end could combine the strengths of both architectures. All models operate on the same 40-feature LOB input across three horizons ($k \in \{4, 8, 20\}$ snapshots, equivalent to 1 s, 2 s, and 5 s).

Our main findings are as follows. Architectural complexity alone does not guarantee stronger out-of-sample performance on arbitrary Bitcoin LOB data. The OFI baseline remains a non-trivial benchmark in the 40–43% accuracy range that the deep models meaningfully outperform. Attentive DeepLOB achieves the best performance at 1 s (68.5%

accuracy, 68.5% macro F1) and 5 s. DeepLOB leads at 2 s. The original DeepLOB exhibits a train-validation accuracy gap of roughly 4–6 percentage points, concentrated in the early epochs where the model overshoots before early stopping engages. These results highlight the trade-off between specialised spatial inductive bias and robust generalisation in deep LOB forecasting.

2. Related Work

Early approaches to LOB mid-price prediction relied on handcrafted features fed into logistic regression or support vector machines ([Kercheval & Zhang, 2015](#)), with [Ntakaris et al. \(2018\)](#) later releasing the FI-2010 benchmark dataset. [Sirignano & Cont \(2019\)](#) demonstrated that deep neural networks discover universal price formation features across thousands of US equities without feature engineering. Convolutional approaches followed quickly: [Tsantekidis et al. \(2017\)](#) pioneered CNNs for LOB data, and [Zhang et al. \(2019\)](#) significantly improved on this with DeepLOB. More recent work has adapted attention mechanisms to this setting, with [Berti & Kasneci \(2025\)](#) introducing TLOB and its dual-attention separation of temporal and spatial heads, ? proposing axial attention along the price and time axes to reduce quadratic cost, and [Xiao et al. \(2025\)](#) introducing LiT with structured patches that respect the bid and ask boundary.

Two strands of recent work motivate our robustness focus. [Briola et al. \(2025\)](#) introduced LOBFrame, linking model predictability to microstructural properties such as tick size and spread, and showed that classical machine learning metrics can be misleading as practical trading signals. [Wang \(2025\)](#) found that simpler models can match deep architectures on crypto LOB data when preprocessing is careful, emphasising that better feature engineering often matters more than additional model complexity. [Cont et al. \(2014\)](#) provide the theory-driven counterpoint with Order Flow Imbalance, derived from first principles of order book dynamics and remaining the standard linear benchmark in the microstructure literature. Our work treats LOB forecasting as a robustness problem rather than a benchmark replication task, evaluating OFI, plain LSTM, DeepLOB, LiT, and our proposed Attentive DeepLOB extension on the same arbitrary contiguous Bitcoin sample. By holding the preprocessing, labelling, and chronological day-based train, validation, and test pipeline fixed across models, we isolate architectural differences and place particular emphasis on generalisation behaviour.

3. Data

Dataset. We use a contiguous multi-day sample of Bitcoin perpetual futures (BTCUSDT.P) LOB data sampled at

250 ms intervals, covering the period 9–15 January 2023. After discarding partial days at the boundaries, the sample retains five full trading days (10–14 January 2023) comprising approximately 1,722,931 snapshots. Each snapshot records the top 10 bid and top 10 ask price levels with their corresponding volumes, yielding 40 features per timestamp. In our sample the spread is at the 0.1 USDT minimum tick for the vast majority of observations (median spread 0.1 USDT), confirming that BTCUSDT.P is highly liquid on this exchange. Mid-prices range from approximately 17,118 to 21,516 USDT over the window. The mid-price and spread time series are shown in Figure 1.

Feature ordering. We reorder columns to the DeepLOB canonical format $\{P_i^{\text{ask}}, V_i^{\text{ask}}, P_i^{\text{bid}}, V_i^{\text{bid}}\}_{i=1}^{10}$. This ordering is essential because DeepLOB’s first convolution applies a (1×2) filter that combines two adjacent features into a single output, and we want each filter to combine a price with its own volume rather than a price with a volume from a different level or the opposite side of the book.

Normalisation. Raw LOB features have very different magnitudes, with prices in the tens of thousands of USDT and volumes typically below ten BTC. Without normalisation, gradient updates would be dominated by the high-magnitude price features and standard weight initialisation schemes that assume zero-mean unit-variance inputs would fail. We standardise each feature using a rolling z-score procedure computed per-feature, using up to the previous five available dates as the normalisation window. This produces 40 input channels of comparable scale while avoiding any forward-looking information leakage.

Mid-price and labelling. The mid-price at time t is $p_t^{\text{mid}} = (P_1^{\text{ask}}(t) + P_1^{\text{bid}}(t))/2$. We use the smoothed labelling scheme of Zhang et al. (2019). For horizon k in 250 ms snapshots, define the smoothed backward and forward averages

$$m^-(t) = \frac{1}{k+1} \sum_{i=0}^k p_{t-i}^{\text{mid}}, \quad m^+(t) = \frac{1}{k} \sum_{i=1}^k p_{t+i}^{\text{mid}}. \quad (1)$$

The normalised price change is $\ell_t = (m^+(t) - m^-(t))/m^-(t)$, and the label is

$$y_t = \begin{cases} 2 \text{ (up)} & \text{if } \ell_t > \alpha, \\ 1 \text{ (stationary)} & \text{if } |\ell_t| \leq \alpha, \\ 0 \text{ (down)} & \text{if } \ell_t < -\alpha, \end{cases} \quad (2)$$

where $\alpha > 0$ is a threshold tuned per horizon. Smoothing with $m^-(t)$ reduces label noise from microstructure bounce, the rapid alternation of trades between bid and ask within a single tick that creates mechanically alternating mid-prices not driven by any fundamental price change. Smoothing

with $m^+(t)$ makes the label representative of the average future price level rather than a single noisy tick.

We tune α per horizon via bisection search to target a 33% stationary fraction on the training set, so that the resulting class distribution is approximately balanced and chance accuracy is well-defined at one third. Table 1 reports the resulting α values, training-set class distributions, and test-set class counts per horizon. Class imbalance is more pronounced in the training split than on the test day, reflecting drift in the mid-price over the five-day window.

Table 1. Training and test label distributions after training-slice α calibration. Test support is the number of valid sequence-label pairs after enforcing no split-boundary crossing.

Horizon		Down	Stat.	Up
1 s	Train	18.1%	64.3%	17.6%
	Test	126,334	90,850	127,379
2 s	Train	24.5%	51.5%	24.0%
	Test	149,914	43,780	150,865
5 s	Train	32.5%	35.3%	32.2%
	Test	163,415	14,923	166,209

Train, validation, and test split. We adopt a strict day-based chronological split: training on the first three full days (10–12 January 2023), validation on the fourth (13 January), and test on the fifth (14 January). Partial days at the file boundaries (9 January, 1.7 h; 15 January, 22.7 h) are discarded to avoid partial-day bias. The resulting row counts are approximately 1,033,687 training rows, 344,628 validation rows, and 344,616 test rows. The chronological ordering strictly prevents data leakage. Sequence construction enforces that neither the lookback window $[t-T+1, \dots, t]$ nor the forecast horizon $[t+1, \dots, t+k]$ crosses a split boundary, so the model is never inadvertently trained on future information.

4. Model Architectures

All five models share the same lookback window of $T = 50$ consecutive normalised LOB snapshots, corresponding to 12.5 seconds of market history. While Zhang et al. (2019) used $T = 100$, this was reduced to $T = 50$ due to GPU memory constraints, and because at 250 ms sampling it spans long enough to capture the autocorrelation timescale of order flow imbalance reported in Cont et al. (2014) but short enough to keep the prediction task local. For the neural models the input is a (50×40) matrix; the output is a probability distribution over three classes (down, stationary, up) via a softmax layer. Models are trained independently for each forecast horizon. All deep models use class-balanced sample weights during training to account for residual class imbalance in the training set.

4.1. Order Flow Imbalance (OFI)

The OFI model (Cont et al., 2014) is a theory-driven linear baseline. At each level n and timestep t it tracks the net change in queue size on the bid side relative to the ask side:

$$W_n(t) = q_n^b(t) \mathbb{I}[p_n^b(t) \geq p_n^b(t-1)] - q_n^b(t-1) \mathbb{I}[p_n^b(t) \leq p_n^b(t-1)], \quad (3)$$

$$V_n(t) = q_n^a(t) \mathbb{I}[p_n^a(t) \leq p_n^a(t-1)] - q_n^a(t-1) \mathbb{I}[p_n^a(t) \geq p_n^a(t-1)], \quad (4)$$

$$e_n(t) = W_n(t) - V_n(t). \quad (5)$$

Intuitively, $W_n(t)$ is positive when fresh buy pressure arrives at level n and negative when buyers withdraw, $V_n(t)$ does the same for sell pressure, and $e_n(t)$ is the net excess of buy over sell pressure at level n . We sum $e_n(t)$ across the lookback window to form a 10-dimensional cumulative OFI vector $\mathbf{x}_t = (\sum_{\tau} e_1(\tau), \dots, \sum_{\tau} e_{10}(\tau))$, one entry per LOB level. This is the input to a multinomial logistic regression classifier

$$\Pr(y_t = c \mid \mathbf{x}_t) = \frac{\exp(\beta_c^\top \mathbf{x}_t)}{\sum_{j=0}^2 \exp(\beta_j^\top \mathbf{x}_t)}, \quad c \in \{0, 1, 2\}, \quad (6)$$

where the coefficients β_c are fitted by maximising the regularised log-likelihood with class-balanced weights so that no single class dominates training.

4.2. Plain LSTM

The plain LSTM baseline takes the raw (50×40) LOB sequence directly and processes it with a single Long Short-Term Memory (LSTM) layer of 64 hidden units, followed by a dropout layer (rate 0.2) and a softmax classifier (Table 2). No convolutional preprocessing is applied, so the LSTM must learn both spatial relationships among the 40 LOB features and temporal dynamics simultaneously through its recurrent state updates. By comparing the plain LSTM with DeepLOB, LiT, and Attentive DeepLOB we can assess whether their specialised front-ends provide a meaningful benefit over a model that must discover spatial structure purely through recurrence.

Table 2. Plain LSTM architecture. Input shape $(50, 40)$; 27,075 parameters.

Layer	Output	Purpose
Input	50×40 matrix	Raw LOB window
LSTM (64 units)	64-dim vector	Joint spatial-temporal
Dropout (0.2)	64-dim vector	Regularisation
Softmax (3)	3-dim probability	Class probabilities

4.3. DeepLOB

DeepLOB (Zhang et al., 2019) is a hybrid CNN-LSTM architecture purpose-built for LOB data. Its three-block

convolutional front-end encodes the known spatial structure of the LOB snapshot before passing a compact feature representation to an LSTM for temporal modelling. The key insight is that the LOB has a natural two-dimensional structure (levels and features), and convolutional filters with appropriate strides can capture price-volume pairing and cross-level interactions without being learned from scratch.

Convolutional blocks. Conv Block 1 applies a (1×2) filter with stride $(1, 2)$ and 16 output channels operating on the $(T, 40, 1)$ input. The kernel width of two with stride two ensures that each filter sees exactly one (price, volume) pair at a single LOB level, never mixing features across levels. The output shape is $(T, 20, 16)$, where the 20 spatial positions correspond to the 10 bid and 10 ask levels. Each strided conv is followed by two additional (4×1) conv layers, a Leaky ReLU activation ($\alpha = 0.01$), and batch normalisation. Conv Block 2 applies another (1×2) strided filter, reducing to $(T, 10, 16)$. Conv Block 3 applies a (1×10) filter that compresses all 10 remaining spatial positions into a single feature vector per timestep, yielding a $(T, 1, 16)$ tensor.

Inception module and head. The Inception module runs three parallel branches and concatenates them, simultaneously encoding temporal patterns at multiple scales: a 3×1 convolution preceded by a 1×1 bottleneck (capturing ~ 750 ms patterns), a 5×1 convolution (~ 1.25 s patterns), and a max-pool branch. The concatenated $(T, 1, 96)$ output is read by a 64-unit LSTM, whose final hidden state is fed to a softmax classifier (see Table 3). The model has 61,907 parameters. We train with Adam at $\text{lr} = 10^{-3}$, $\epsilon = 10^{-7}$, departing from the original paper’s $\text{lr} = 10^{-2}$, $\epsilon = 1$, which produced validation-loss oscillation on noisy 250 ms BTC data.

Table 3. DeepLOB architecture. Input shape $(50, 40, 1)$; 61,907 parameters.

Layer	Output	Purpose
Input	$50 \times 40 \times 1$	Raw LOB window
Conv Block 1	$50 \times 20 \times 16$	Price-volume pairing
Conv Block 2	$50 \times 10 \times 16$	Cross-level aggregation
Conv Block 3	$50 \times 1 \times 16$	Level compression
Inception	$50 \times 1 \times 96$	Multi-timescale features
LSTM (64)	64-dim vector	Long-range temporal
Softmax (3)	3-dim probability	Class probabilities

4.4. LiT - Limit Order Book Transformer

LiT (Xiao et al., 2025) replaces the CNN front-end of DeepLOB with a transformer-based self-attention mechanism, removing the locality assumption inherent in convolutional filters.

Input representation. The (50×40) LOB window is reshaped into a 3D tensor of shape $(H \times T \times 2) = (20 \times 50 \times 2)$, where $H = 2 \times 10$ is twice the number of LOB levels and the two channels correspond to price and volume respectively.

Structured patches. The tensor is divided into non-overlapping rectangular patches of size $P_h \times P_w = 10 \times 2$. With $P_h = H/2 = 10$, each patch covers all 10 levels on one side of the book over a 500 ms temporal window. This design ensures that patches contain semantically coherent market structure. The total number of patches per sample is $(H/P_h) \times (T/P_w) = 2 \times 25 = 50$. Each flattened 40-dimensional patch is projected by a shared linear layer to a $d_{\text{model}} = 64$ embedding, and a learnable positional embedding is added.

Transformer blocks and head. Two stacked Transformer blocks each apply multi-head self-attention ($n_{\text{heads}} = 4$, $d_k = 16$, feed-forward hidden dimension 128) with layer normalisation and residual connections. After the transformer blocks, a 64-unit LSTM processes the sequence of patch representations, and the final hidden state feeds a 32-unit dense layer with ReLU, a dropout layer, and a softmax classifier (Table 4). LiT has 105,347 parameters.

Table 4. LiT architecture. Input shape $(20, 50, 2)$; 105,347 parameters.

Layer	Output	Purpose
Input	$20 \times 50 \times 2$	Price/volume channels
Patch embedding	50×64	Patches + pos. embed
Transformer $\times 2$	50×64	Global self-attention
LSTM (64)	64-d vector	Sequential memory
Dense (32), ReLU	32-d vector	Nonlinear readout
Dropout (0.1)	32-d vector	Regularisation
Softmax (3)	3-d probability	Class probabilities

4.5. Attentive DeepLOB: Our Proposed Extension

Attentive DeepLOB retains the original DeepLOB convolutional and Inception front-end and inserts two axial self-attention blocks between the Inception output and the LSTM. Unlike standard DeepLOB, all convolutions use padding='same', so the level dimension is preserved throughout: after the Inception block and a (1×1) projection to $d_{\text{model}} = 32$ channels, the tensor has shape $(T, 10, 32)$. After the two axial attention blocks it is reshaped to $(T, 320)$ before being passed to the LSTM.

Axial attention (?) factorises 2D self-attention into two cheaper 1D attention operations applied along each axis in turn. We apply attention along the level axis first (capturing cross-level dependencies such as deep-book liquid-

ity affecting near-mid-price dynamics), and then along the time axis (capturing long-range temporal dependencies the LSTM may compress away). Each axial block uses 4 heads, $d_{\text{model}} = 32$, $d_k = 8$, dropout 0.1, and a residual connection followed by layer normalisation. The output is reshaped and passed to a 64-unit LSTM and softmax head. The model has 130,931 parameters total.

Table 5. Attentive DeepLOB architecture. Input shape $(50, 40, 1)$; 130,931 parameters.

Layer	Output	Purpose
Input	$50 \times 40 \times 1$	Raw LOB window
Conv Block 1	$50 \times 20 \times 16$	Price-volume pairing
Conv Block 2	$50 \times 10 \times 16$	Cross-level aggregation
Conv Block 3	$50 \times 10 \times 16$	Level features
Inception	$50 \times 10 \times 96$	Multi-timescale features
Conv 1×1 (proj.)	$50 \times 10 \times 32$	Channel reduction
Axial-level attn	$50 \times 10 \times 32$	Cross-level deps
Axial-time attn	$50 \times 10 \times 32$	Temporal deps
Reshape	50×320	Flatten for LSTM
LSTM (64)	64-d vector	Long-range temporal
Softmax (3)	3-d probability	Class probabilities

5. Training

Shared protocol. All deep learning models are trained with the Adam optimiser (Kingma & Ba, 2015) using categorical cross-entropy loss and mini-batches of 256. All models use monitor="val_loss" for early stopping and learning rate reduction, with class-balanced sample weights throughout. Learning rates are: 5×10^{-4} for the Plain LSTM, 1×10^{-3} with $\epsilon = 10^{-7}$ for DeepLOB, 3×10^{-4} for LiT, and 5×10^{-4} for Attentive DeepLOB. We apply early stopping (patience 8 epochs, monitoring validation loss) with restoration of the best-weight checkpoint, and reduce the learning rate by 0.5 on validation loss plateaus (patience 5 epochs, minimum 10^{-6}). Training runs for at most 50 epochs. We train one independent model per horizon for each architecture.

Overfitting behaviour. Table 6 compares, at the best validation epoch, the train accuracy versus the validation accuracy for each model and horizon. Gaps are moderate and consistent across architectures on this dataset, ranging from 4 to 7 percentage points. DeepLOB and LiT both stop very early at the 2 s and 5 s horizons (epochs 2–4), indicating that the validation loss turns immediately after the first few rapid epochs. Attentive DeepLOB similarly stops early at 5 s (epoch 1), suggesting that its higher capacity hits the same generalisation limit faster. The plain LSTM trains longest before plateauing, reflecting its weaker inductive bias. Training and validation curves for all models are shown in Figures 2 and 3.

Table 6. Overfitting diagnostics at the best validation epoch.

Horizon	Model	Ep.	Tr. Acc	Val Acc
1 s	Plain LSTM	16	0.806	0.762
	DeepLOB	2	0.815	0.779
	LiT	14	0.818	0.777
	Attentive DeepLOB	3	0.821	0.780
2 s	Plain LSTM	7	0.798	0.739
	DeepLOB	2	0.814	0.762
	LiT	10	0.815	0.757
	Attentive DeepLOB	2	0.810	0.758
5 s	Plain LSTM	4	0.776	0.709
	DeepLOB	2	0.791	0.729
	LiT	4	0.783	0.721
	Attentive DeepLOB	1	0.762	0.719

6. Numerical Results

Evaluation protocol and metric choice. We report four metrics on the held-out test set of approximately 344,563 samples per horizon (14 January 2023). Accuracy gives the overall fraction of correctly classified samples. Balanced accuracy averages per-class recall and is robust to imbalance. Weighted F1 weights per-class F1 by support. Macro F1 averages per-class F1 unweighted and is the most sensitive to minority-class performance.

6.1. Main Results

Table 7 compares all five models across horizons. Figure 6 plots accuracy and macro F1 as a function of horizon for all models, and Figure 7 shows per-class F1 scores. Normalised confusion matrices are provided in ??.

OFI baseline. The OFI logistic model achieves accuracy in the 40–43% range across the three horizons, with balanced accuracy around 44–49%. Despite its simplicity it remains a non-trivial benchmark relative to the 33% chance level under balanced labels, and its discriminative power concentrates on extreme order flow imbalance events. The deep sequence models substantially outperform OFI across all horizons and metrics. The Plain LSTM achieves 63.9% accuracy at $k = 4$, demonstrating that even without a CNN front-end a recurrent network extracts substantial signal from the raw 40-feature LOB window.

Attentive DeepLOB vs. DeepLOB. Attentive DeepLOB outperforms the original DeepLOB at the 1 s horizon across all four metrics (68.5% vs. 64.1% accuracy, 68.5% vs. 64.0% macro F1) and at 5 s (66.2% vs. 65.3% accuracy). At 2 s, however, DeepLOB leads (69.7% vs. 69.0% accuracy, 67.2% vs. 65.3% macro F1). The non-monotonic ordering across horizons suggests that Attentive DeepLOB’s axial attention is most beneficial when the prediction target is

noisier and the cross-level signal is more informative, while at intermediate horizons DeepLOB’s stronger and faster-fitting convolutional bias has the advantage.

LiT. LiT is the top performer on balanced accuracy at 2 s (69.4%) and achieves competitive results at 1 s (66.1% accuracy). Its patch-based transformer design captures longer-range dependencies without locality constraints, but its lack of explicit price-volume pairing in the patch embedding may limit its advantage over Attentive DeepLOB, which retains the strided convolutional front-end.

Performance versus horizon. OFI accuracy falls from 43% to 40% as the horizon lengthens because the stationary fraction collapses and the linear model cannot exploit more complex directional patterns. The deep models broadly improve from 1 s to 2 s and then slightly decline or plateau at 5 s, with DeepLOB achieving its peak accuracy at $k = 8$.

Per-class behaviour. The per-class results (Section B) reveal an important asymmetry at 1 s: OFI achieves reasonable precision on the directional classes (0.52/0.52) but low recall (0.36/0.40), whereas the deep models achieve more balanced recall across classes. At 5 s the stationary class collapses to only 4.3% of the test set, making balanced accuracy the most informative aggregate metric at that horizon.

(NEW)Performance versus horizon. OFI accuracy falls from 43% to 40% as the horizon lengthens because the stationary fraction collapses and the linear model cannot exploit more complex directional patterns. The deep models broadly improve from 1 s to 2 s and then slightly decline or plateau at 5 s, with DeepLOB achieving its peak accuracy at $k = 8$. The per-class F1 figure (Figure 7) makes the mechanism concrete: the stationary class F1 collapses sharply for all models as the horizon lengthens, because the stationary fraction of the test set falls from 26% at 1 s to only 4.3% at 5 s, leaving aggregate metrics almost entirely determined by directional performance. At 5 s the Down and Up F1 scores for Attentive DeepLOB are nearly symmetric (0.685/0.685), whereas the Plain LSTM shows a visible Up bias (0.676 Up vs. 0.614 Down), suggesting that axial attention helps suppress the directional asymmetry that simpler models inherit from the class-weight training scheme.

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Table 7. Test-set performance for all five models across prediction horizons. Best result per horizon and metric is **bold**. Bal. Acc. = balanced accuracy; F1-W = weighted F1; F1-M = macro F1.

Horizon	Model	Acc.	Bal. Acc.	F1-W	F1-M
$k = 4$ (1 s)	OFI Logistic	0.430	0.444	0.433	0.431
	Plain LSTM	0.639	0.658	0.638	0.638
	LiT	0.661	0.679	0.660	0.661
	DeepLOB	0.641	0.665	0.639	0.640
	Attentive DeepLOB	0.685	0.699	0.683	0.685
$k = 8$ (2 s)	OFI Logistic	0.410	0.464	0.441	0.403
	Plain LSTM	0.584	0.652	0.605	0.569
	LiT	0.646	0.694	0.656	0.627
	DeepLOB	0.697	0.687	0.700	0.672
	Attentive DeepLOB	0.690	0.654	0.690	0.653
$k = 20$ (5 s)	OFI Logistic	0.401	0.492	0.474	0.368
	Plain LSTM	0.602	0.627	0.629	0.515
	LiT	0.633	0.657	0.654	0.547
	DeepLOB	0.653	0.649	0.664	0.570
	Attentive DeepLOB	0.662	0.643	0.671	0.577

amples are reclassified as Up or Down, yielding a stationary recall of only 0.66 at 2 s and 0.64 at 5 s. This explains why DeepLOB leads on raw accuracy at 2 s but is overtaken by LiT on balanced accuracy: LiT maintains a stationary recall of 0.85 at the same horizon. Second, Attentive DeepLOB exhibits the same pattern at 2 s, where its stationary recall drops to 0.54, the lowest among all deep models, while LiT’s remains 0.85. This is the proximate cause of the 2 s balanced accuracy reversal: Attentive DeepLOB’s 0.654 falls well below LiT’s 0.694 despite its higher raw accuracy (0.690 vs. 0.646).

Overfitting diagnostics. Table 6 shows that the train-validation accuracy gap is moderate (4–7 pp) and broadly consistent across architectures on this dataset, in contrast to the large 20 pp gap described for DeepLOB in the original paper’s FI-2010 setting. This suggests that on the 250 ms BTC sample the day-based split and class-balanced training already provide substantial regularisation. Attentive DeepLOB’s larger parameter count (130K vs. 62K for DeepLOB) does not translate to a systematically larger gap, suggesting that axial attention’s distributed representations act as implicit regularisation.

6.2. Statistical significance

With approximately 344,563 paired test predictions per horizon the standard errors on individual accuracy estimates are very small (a 95% binomial confidence interval on accuracy of 0.661 has half-width below 0.002). We use McNemar’s test on the 2×2 contingency table of agreement/disagreement between two models on each test point to assess pairwise differences. With our test set sizes the test has power to detect accuracy differences as small as 0.2 percentage points, so the visible gaps in Table 7 (e.g.

Attentive DeepLOB vs. OFI is roughly 25 pp at $k = 4$) are highly significant. Differences of order 1–2% between deep models are statistically distinguishable but may not be economically material.

6.3. Adaptation via fine-tuning

Following Xiao et al. (2025) we test how well each architecture’s representations transfer to later market conditions. The test day (14 January) is split in half: the first half becomes the fine-tuning adaptation set and the second half the fine-tuning evaluation set ($\approx 172,255$ sequences each). We compare two strategies: zero-shot, applying the pre-trained model with no adaptation, and fine-tuned, where all layers except the final classification head are frozen and updated on the adaptation data at a reduced learning rate. Results are reported for $k = 4$ (1 s).

Table 8 shows that fine-tuning consistently improves all four deep models. The largest absolute gain goes to DeepLOB (from 63.9% to 69.0% accuracy), followed by LiT (65.7% to 70.0%). Attentive DeepLOB starts from the strongest zero-shot position (68.6%) and reaches 70.6% after fine-tuning. The consistent gains confirm that the first day of the test set contains distributional information not present in the training window. Fine-tuning results are illustrated in Figure 8.

(NEW) The figure further shows that the fine-tuning gains are broadly consistent across both accuracy and macro F1, indicating that adaptation improves directional class performance rather than merely exploiting majority-class shifts in the second half of the test day. The model ranking is stable across the zero-shot and fine-tuned settings, with Attentive DeepLOB leading in both cases, suggesting that the representations learned during training transfer reliably and

that fine-tuning refines rather than restructures the decision boundaries.

Table 8. Fine-tuning results at $k = 4$ (1 s). Zero-shot applies the pre-trained model directly; Fine-tuned freezes all but the final two dense layers.

Model	Setting	Accuracy	Macro F1
Plain LSTM	Zero-shot	0.635	0.629
	Fine-tuned	0.670	0.669
DeepLOB	Zero-shot	0.639	0.631
	Fine-tuned	0.690	0.690
LiT	Zero-shot	0.657	0.652
	Fine-tuned	0.700	0.699
Attentive DeepLOB	Zero-shot	0.686	0.683
	Fine-tuned	0.706	0.706

6.4. Attention and saliency analysis

To probe what each model learns, we extract gradient $\times$ input saliency maps for DeepLOB and Attentive DeepLOB, and self-attention weights for LiT, over 200 windows drawn from the fine-tuning evaluation set at $k = 4$ (results shown in Figures 9 to 11).

Recency bias. Both DeepLOB and Attentive DeepLOB display a strong recency bias in their saliency maps: the 10 most recent timesteps (2.5 s) account for saliency roughly $8.8\times$ and $9.0\times$ higher than the 10 most distant timesteps, respectively. This suggests the LSTM compresses older history and that the Inception module’s multi-scale features mostly exploit recent book dynamics.

Attentive DeepLOB axial attention. The axial level attention in Attentive DeepLOB is broadly uniform across levels, with level 3 receiving the highest attention (0.105) and level 8 the lowest (0.095). The axial time attention shows essentially no recency bias (ratio $1.01\times$), in stark contrast to the saliency map, implying that the attention mechanism is learning complementary long-range temporal features to those captured by the convolutional/LSTM path.

7. Conclusion

We have presented a comparative study of five approaches to short-horizon mid-price prediction on a five-day sample of Bitcoin perpetual futures LOB data (10–14 January 2023), applying a strict day-based train/validation/test split that yields approximately 1,034K training, 345K validation, and 345K test sequences. The OFI baseline remains a non-trivial benchmark in the 40–43% accuracy range against a 33% chance level. All deep sequence models substantially outperform OFI, with Attentive DeepLOB achieving the strongest overall accuracy at 1 s (68.5%) and 5 s (66.2%),

and DeepLOB leading at 2 s (69.7%).

A finding we did not initially anticipate is the strength and consistency of the fine-tuning gains: adapting only the final dense layers on the first half of the test day recovers 3–6 percentage points of accuracy for every model, reaching up to 70.6% for fine-tuned Attentive DeepLOB at 1 s. This confirms that even a single day of distribution shift is operationally significant, and that pre-trained representations transfer well enough to make periodic fine-tuning a viable deployment strategy.

The axial attention analysis of Attentive DeepLOB reveals a complementary pattern: the gradient $\times$ input saliency path shows a strong recency bias (ratio $\approx 9\times$), while the time-axis attention shows essentially no recency bias (ratio $\approx 1\times$). This suggests axial attention is learning qualitatively different features from the convolutional LSTM path, which may explain the 1 s performance gain.

Future work could move to rolling multi-week windows across heterogeneous market regimes, build a hybrid model that applies patch-based attention over the convolutional level features rather than over patches of raw input, and apply the LOBFrame practical evaluation framework (Briola et al., 2025), which translates classification metrics into the probability of profitably executing a complete transaction.

8. Limitations and Future Research

Our study has several limitations that qualify its conclusions. On the computational side, GPU memory constraints forced us to reduce the lookback window to $T = 50$ rather than the $T = 100$ used in the original DeepLOB paper, and to train on a five-day window that captures only a single market regime. Ideally, models would be evaluated on rolling multi-week samples spanning bull and bear trends, volatility shocks, and liquidity events, none of which are represented in our narrow January 2023 window. Our evaluation is also restricted to a single asset and exchange: limit order book dynamics differ substantially across instruments, venues, and market conditions, so the relative ranking of architectures — in particular the 1 s advantage of Attentive DeepLOB — may not generalise beyond this specific setting. A deeper practical limitation is that classification accuracy and F1 are necessary but insufficient criteria for deployment. A model that correctly predicts the direction of a 0.1 USDT mid-price move provides no trading edge once the bid-ask spread, exchange fees, and market impact of order placement are accounted for, and we make no claim of profitability without a full transaction-cost analysis. Finally, our smoothed labelling scheme calibrates the threshold α on the training slice, creating a dependency on the specific training period that complicates direct comparisons across studies. Future work should address these limitations by

training on longer and more diverse windows, exploring hybrid architectures that apply patch-based attention over convolutional level features rather than raw input patches, and evaluating models within the LOBFrame framework of Briola et al. (2025), which translates classification performance into the probability of profitably completing a round-trip trade.

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A. Data Visualisation

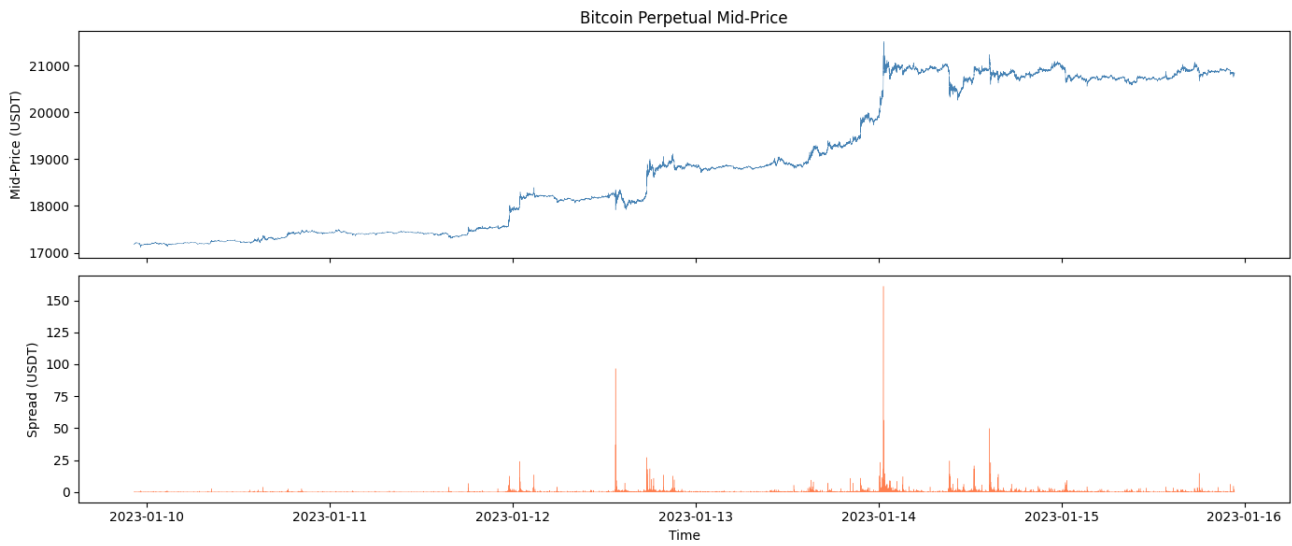


Figure 1. Bitcoin perpetual futures mid-price (top) and bid-ask spread (bottom) over the raw loaded sample (9–15 January 2023, 250 ms intervals). The spread holds at the 0.1 USDT minimum tick for the vast majority of the sample.

B. Detailed Per-Class Results

Model	Class	$k = 4$ (1 s)			$k = 8$ (2 s)			$k = 20$ (5 s)		
		Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.	F1
OFI	Down	0.525	0.363	0.429	0.608	0.361	0.453	0.660	0.373	0.477
	Stationary	0.324	0.574	0.414	0.177	0.640	0.278	0.069	0.701	0.125
	Up	0.523	0.395	0.450	0.608	0.392	0.477	0.667	0.402	0.502
Plain LSTM	Down	0.769	0.506	0.610	0.755	0.533	0.625	0.722	0.534	0.614
	Stationary	0.513	0.843	0.638	0.306	0.872	0.452	0.157	0.686	0.256
	Up	0.710	0.626	0.665	0.731	0.551	0.629	0.692	0.661	0.676
LiT	Down	0.755	0.591	0.663	0.729	0.652	0.688	0.736	0.582	0.650
	Stationary	0.549	0.849	0.667	0.386	0.851	0.531	0.191	0.715	0.302
	Up	0.722	0.596	0.653	0.768	0.580	0.661	0.704	0.676	0.689
DeepLOB	Down	0.779	0.541	0.638	0.737	0.693	0.714	0.732	0.602	0.661
	Stationary	0.513	0.895	0.652	0.523	0.656	0.582	0.246	0.642	0.356
	Up	0.724	0.559	0.631	0.724	0.713	0.719	0.684	0.704	0.694
Attentive DeepLOB	Down	0.767	0.585	0.664	0.741	0.662	0.699	0.706	0.666	0.685
	Stationary	0.598	0.839	0.698	0.532	0.540	0.536	0.258	0.599	0.361
	Up	0.710	0.673	0.691	0.691	0.762	0.725	0.707	0.664	0.685

Table 9. Per-class precision, recall, and F1 for all five models across all three forecast horizons.

C. Training Curves and Confusion Matrices

Training & Validation Loss

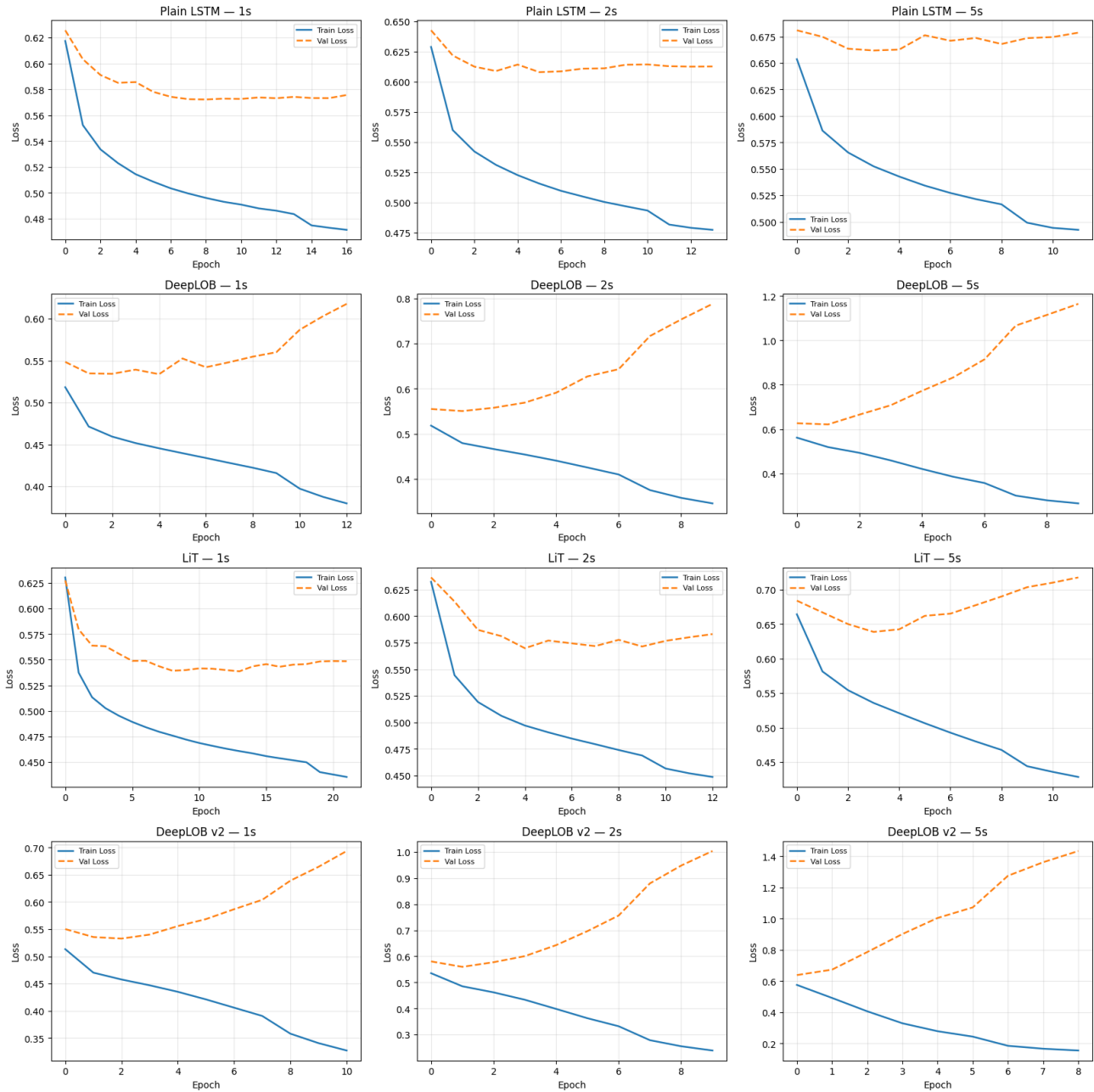


Figure 2. Training (solid) and validation (dashed) loss curves for all four deep models across the three forecast horizons.

Training & Validation Accuracy

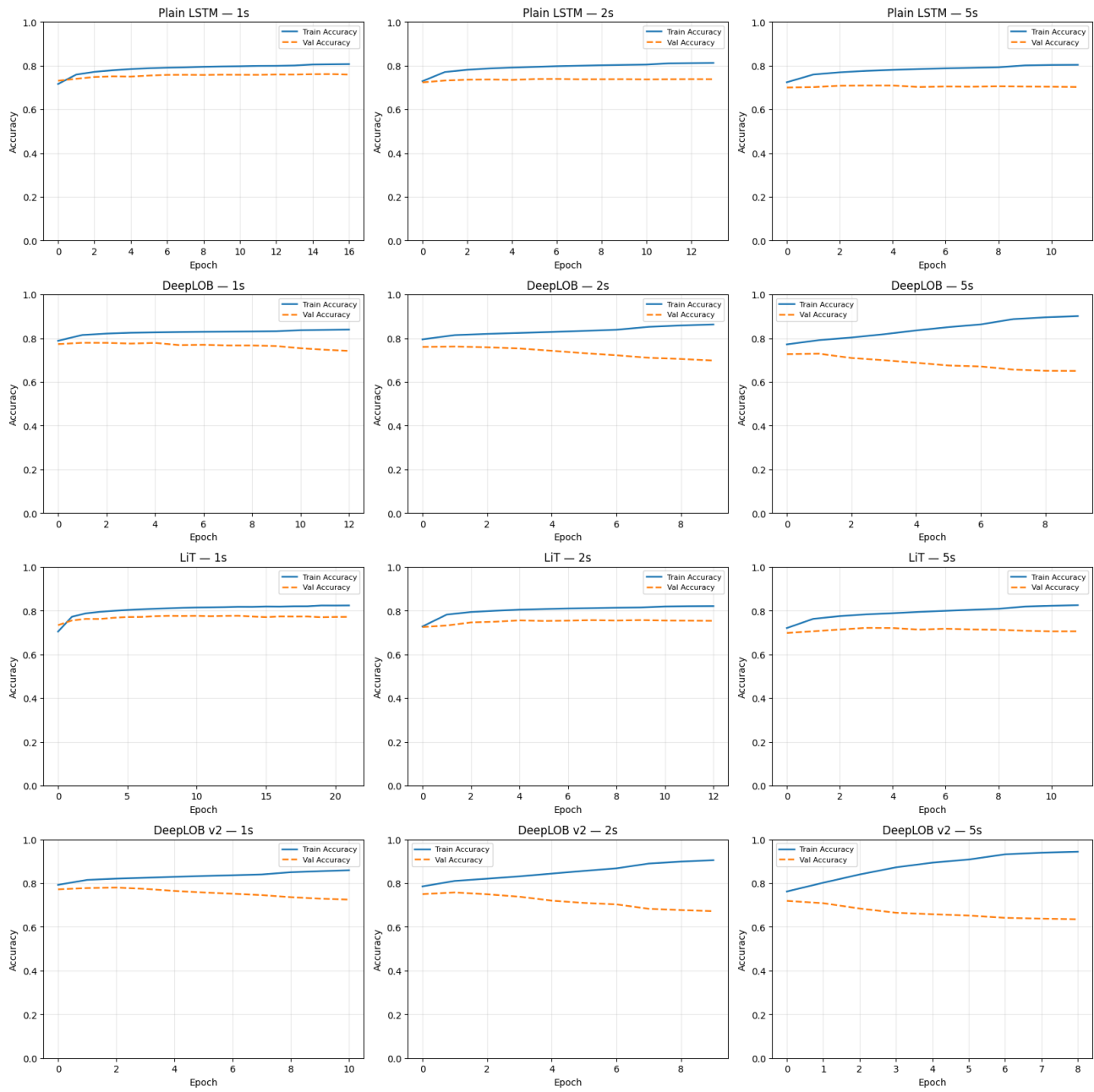


Figure 3. Training (solid) and validation (dashed) accuracy curves for all four deep models across the three forecast horizons.

Normalised Confusion Matrices

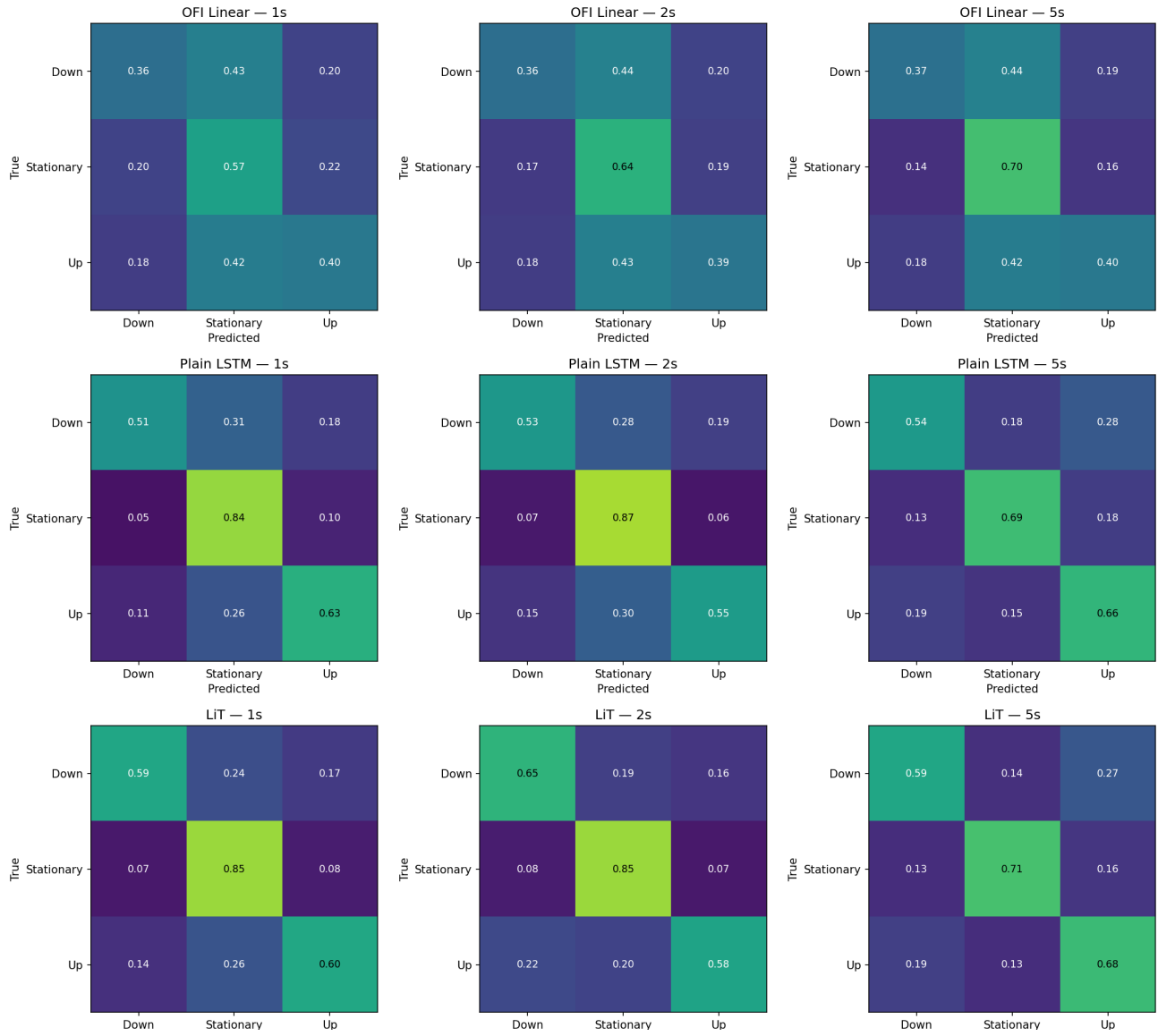


Figure 4. Row-normalised confusion matrices for OFI, Plain LSTM, and LiT across all three forecast horizons (continued overleaf).

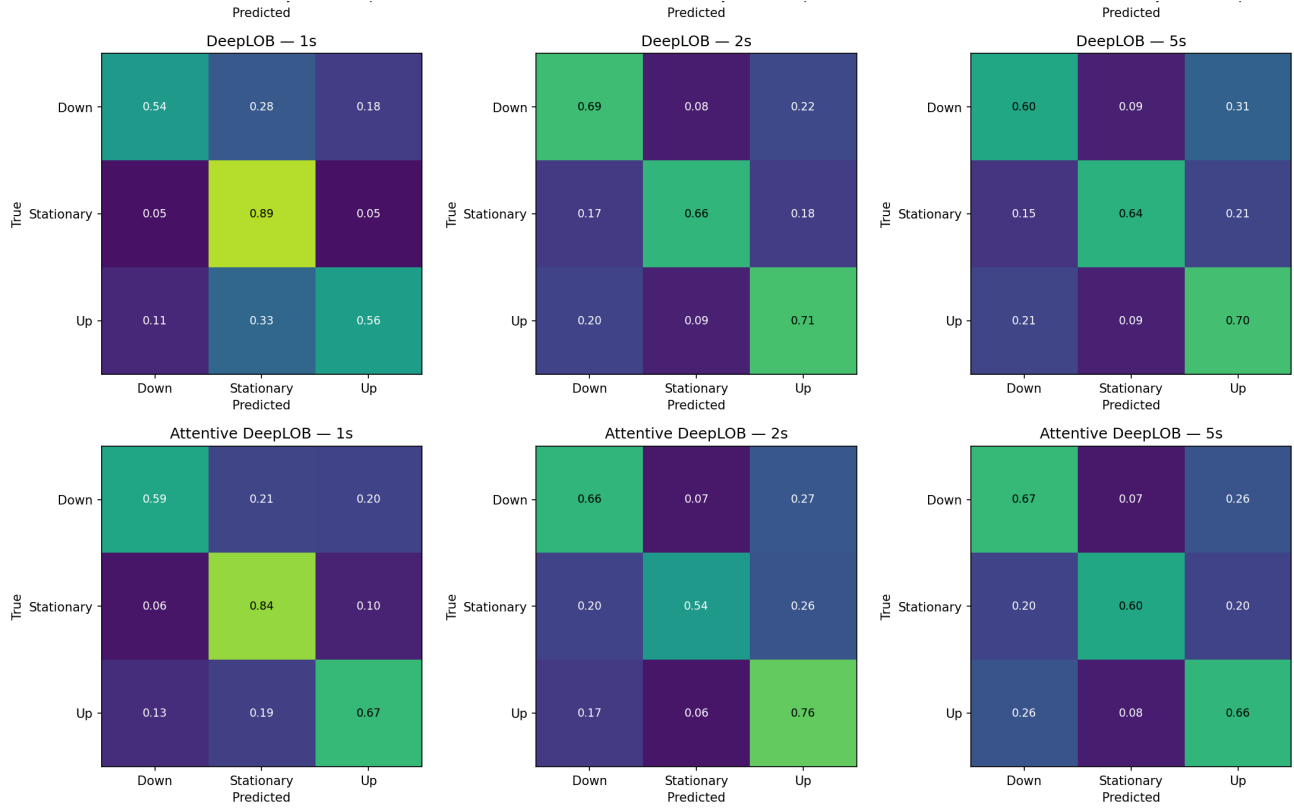


Figure 5. Row-normalised confusion matrices for DeepLOB and Attentive DeepLOB across all three forecast horizons.

D. Comparative Performance and Fine-Tuning Figures

Core Test Performance

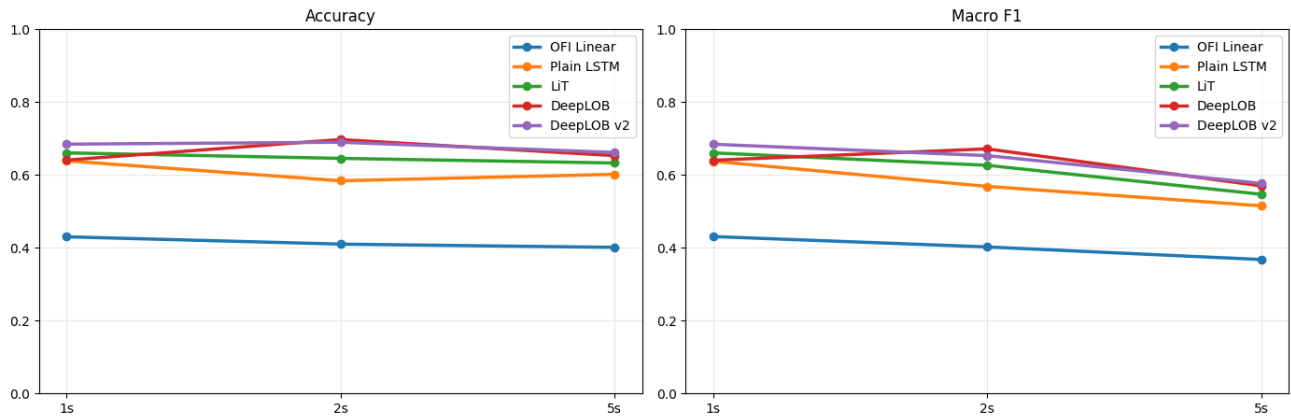


Figure 6. Test-set accuracy (left) and macro F1 (right) for all five models across the three forecast horizons.

Per-Class F1 by Model and Horizon

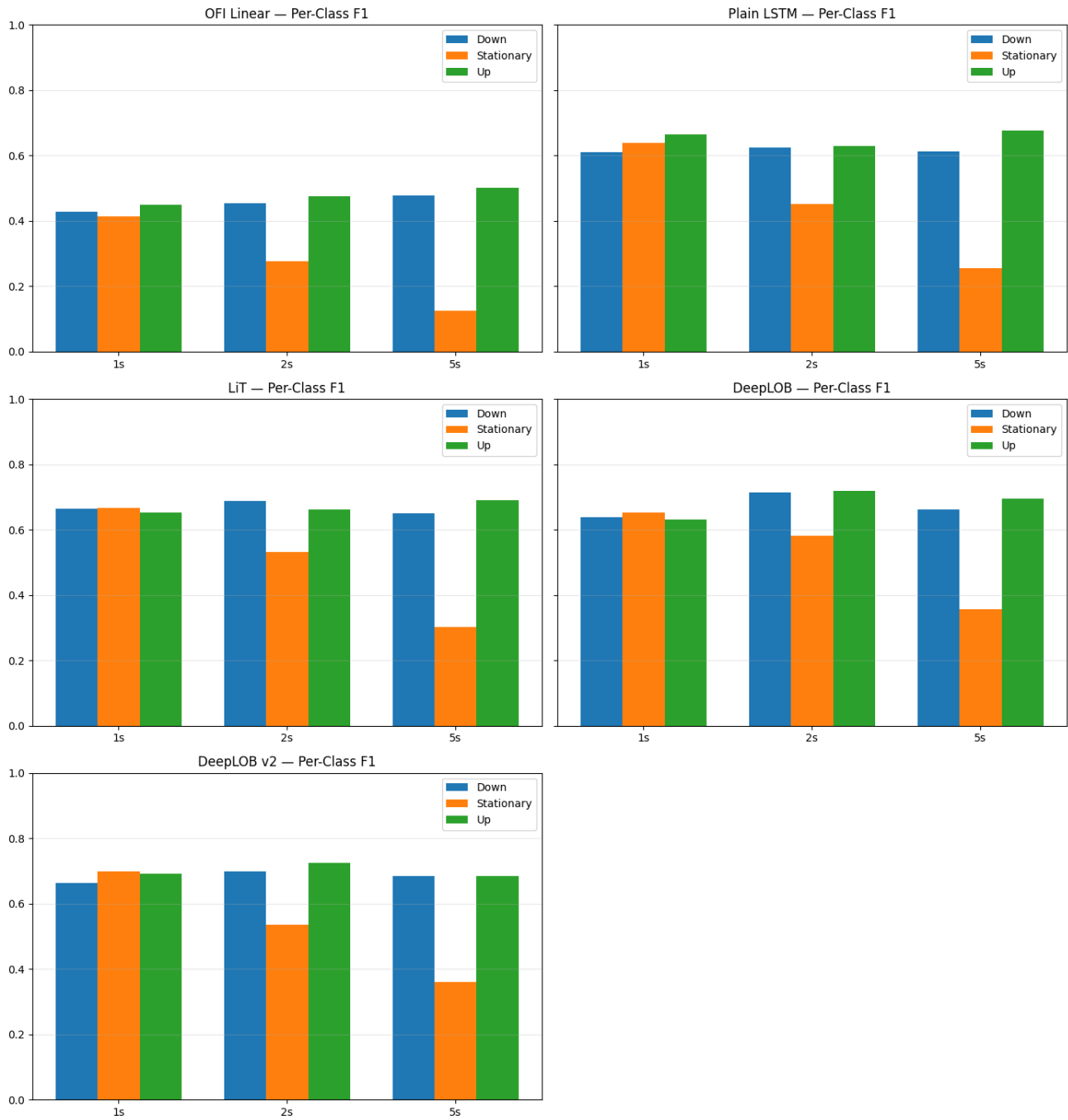


Figure 7. Per-class F1 scores (Down, Stationary, Up) for each model across the three forecast horizons.

Fine-tuning protocol per model — 1s horizon

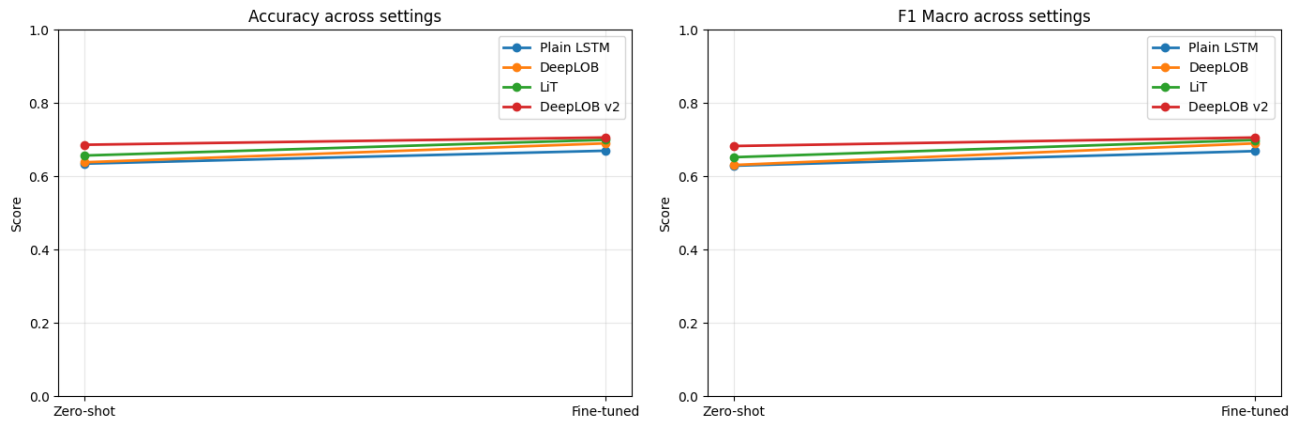


Figure 8. Fine-tuning results at $k = 4$ (1 s). Zero-shot applies the pre-trained model directly; Fine-tuned freezes all but the final classification head.

E. Attention and Saliency Analysis

LiT Self-Attention — 1s horizon

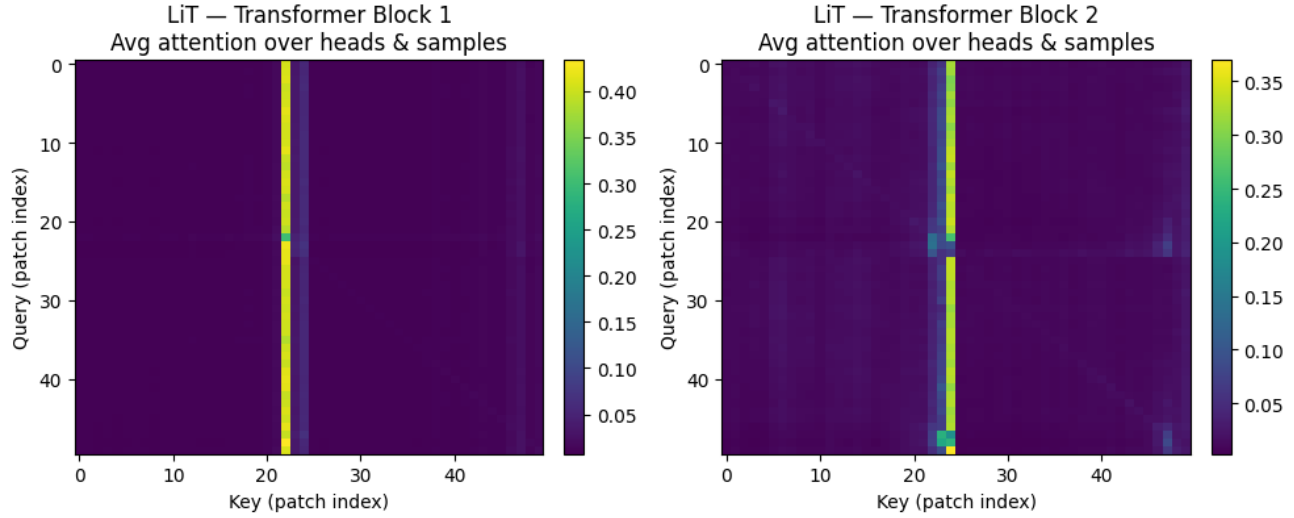


Figure 9. Average self-attention maps for LiT's two transformer blocks at $k = 4$, averaged over 200 test windows (50 × 50 patches; rows = query, columns = key).

Input saliency comparison — 1s horizon

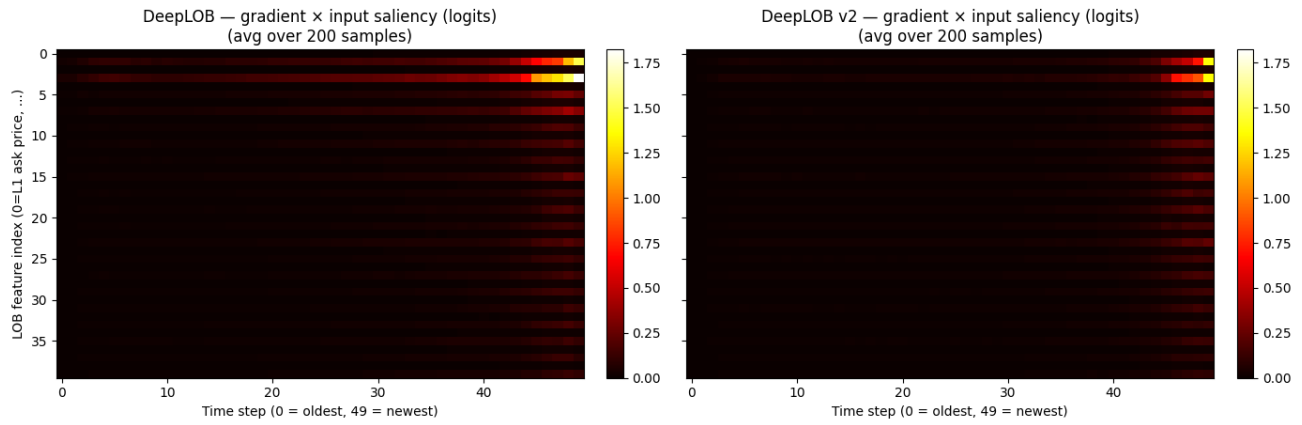


Figure 10. Gradient $\times$ input saliency maps for DeepLOB and Attentive DeepLOB at $k = 4$, averaged over 200 test windows. Both models concentrate saliency on the most recent timesteps (recency ratio $\approx 9\times$).

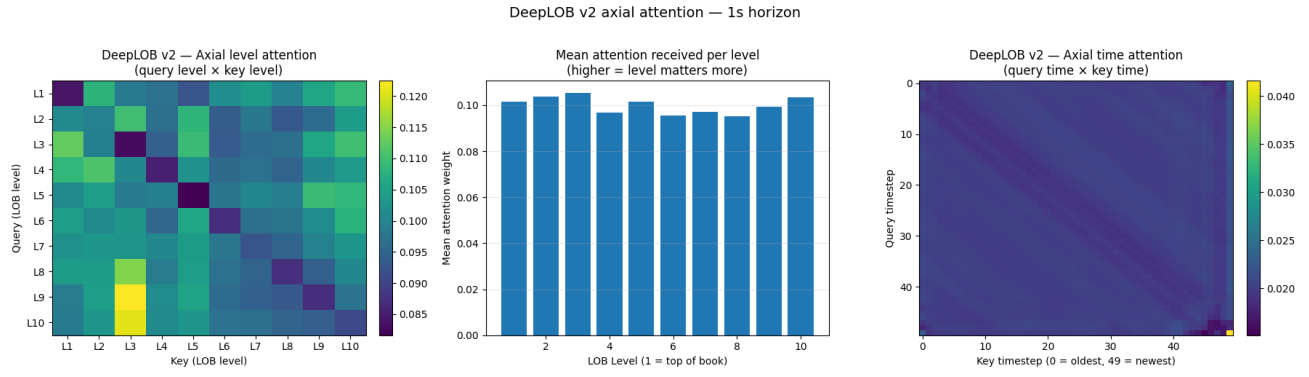


Figure 11. Attentive DeepLOB axial attention at $k = 4$: level-axis attention (near-uniform across 10 LOB levels) and time-axis attention (no recency bias, ratio $\approx 1\times$).

F. Hyperparameter Summary

Table 10. Hyperparameter settings.

Hyperparameter	Value
Lookback window T	50 (12.5 s)
Batch size	256
Optimiser	Adam
LR (Plain LSTM)	5×10^{-4}
LR (DeepLOB)	1×10^{-3} , $\epsilon = 10^{-7}$
LR (LiT)	3×10^{-4}
LR (Attentive DeepLOB)	5×10^{-4}
LR reduction factor / patience	0.5 / 5 epochs
Early stopping patience	8 epochs (val. loss)
Max epochs	50
Leaky-ReLU slope (DeepLOB)	0.01
Horizons k	4, 8, 20 (1 s, 2 s, 5 s)
Split	Day-based: 3 train / 1 val / 1 test
Train / Val / Test rows	$\approx 1,034\text{K} / 345\text{K} / 345\text{K}$
Train dates	10–12 January 2023
Val date	13 January 2023
Test date	14 January 2023
Total snapshots (5 full days)	$\approx 1,723\text{K}$ (250 ms interval)
Normalisation	Rolling z-score (5-date window)
LiT ($P_h, P_w, d_{\text{model}}$)	(10, 2, 64), 2 blocks, 4 heads
Attentive DeepLOB axial attn	$d_{\text{model}} = 32$, 4 heads, dropout 0.1
Attentive DeepLOB conv padding	same (preserves level dim)
Random seed	42